

CEPHALOPODS

Cephalopods are agile hunters in the mollusc phylum. Their sophisticated nervous system enables them to hunt down fast-moving prey.

Cephalopods include the most intelligent of all invertebrates. Many species signal their mood using pigment-containing structures in their skin called chromatophores. The class is organized by number of arms. Squid and cuttlefish have eight arms used for swimming, plus two longer retractable arms, called tentacles, with suckers for gripping prey. Octopus lack tentacles, but have suckers on all eight of their arms. The cephalopod mantle encloses a cavity, which contains the gills. To obtain oxygen, cephalopods take in water through the sides of the mantle. It passes over the

gills and is then expelled through a short funnel. Cephalopods can move backwards quickly by expelling water through the funnel, to propel themselves forwards. Squid, cuttlefish, and some octopus have fins on the sides of the mantle, used for open-water swimming. Most octopus spend their time on the sea floor.

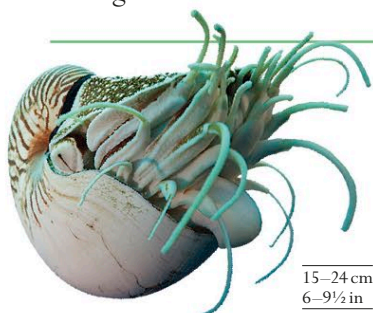
Only the open-ocean nautilus has a coiled shell. In squid the shell is reduced to a pen-like structure that gives some internal support, in cuttlefish to a similar, calcified structure called a cuttlebone. Most species of octopus have lost their shell.

Cephalopods are fast-moving predators that use their arms to grasp prey, and a parrot-like beak to despatch it. Squid catch free-swimming prey, while cuttlefish and octopus take slower, bottom-crawling crustaceans, such as crabs.

PHYLUM	MOLLUSCA
CLASS	CEPHALOPODA
ORDERS	9
FAMILIES	About 50
SPECIES	822



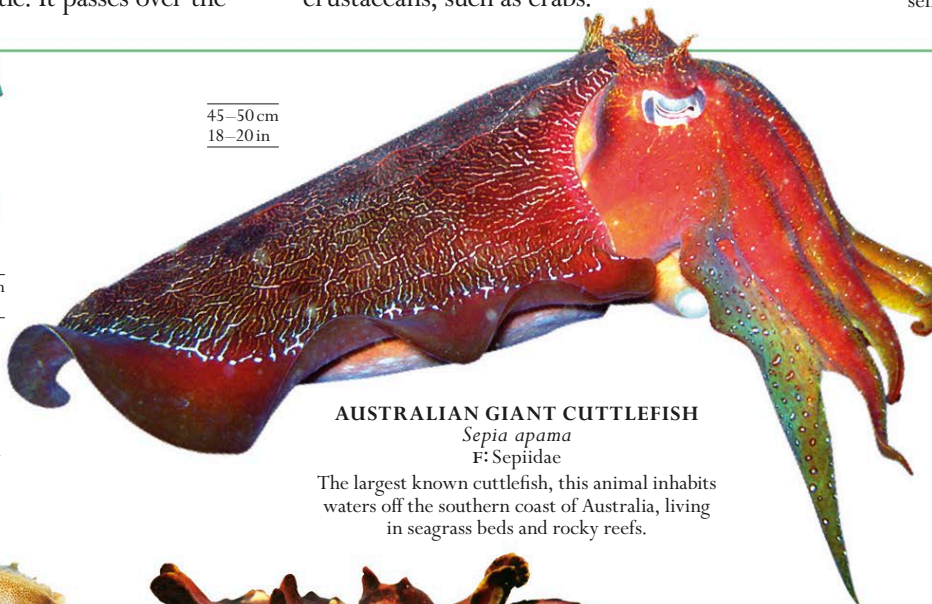
A North Pacific giant octopus discharges black-pigmented ink in self-defence as it jets away from a predator.



15–24 cm
6–9½ in

CHAMBERED NAUTILUS
Nautilus pompilius
F: Nautilidae

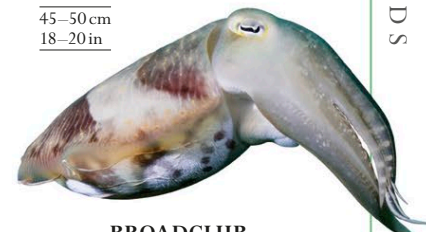
This Indo-Pacific species is the largest and best-known living member of a small group of cephalopods otherwise known only as fossils.



45–50 cm
18–20 in

AUSTRALIAN GIANT CUTTLEFISH
Sepia apama
F: Sepiidae

The largest known cuttlefish, this animal inhabits waters off the southern coast of Australia, living in seagrass beds and rocky reefs.



45–50 cm
18–20 in

BROADCLUB CUTTLEFISH
Sepia latimanus
F: Sepiidae

This large cuttlefish is wide-ranging throughout the Indo-Pacific region. It is an abundant predator on coral reefs, feeding on prawns and shrimps.



40–50 cm
16–20 in

COMMON CUTTLEFISH
Sepia officinalis
F: Sepiidae

Like many other cuttlefish, this species migrates inshore to spawn on muddy sediments. It occurs off the coasts of Europe and South Africa.



6–7 cm
2¼–2¾ in

FLAMBOYANT CUTTLEFISH
Metasepia pfefferi
F: Sepiidae

Unusual among cuttlefish for its habit of using its tentacles to walk on ocean floors, this Indo-Pacific cuttlefish has recently been shown to be poisonous.



WHIP-LASH SQUID
Mastigoteuthis sp.
F: Mastigoteuthidae

Reddish whip-lash squid such as this species hover with broad fins in deep water, waiting for prey with its long tentacles extended.



2–3 cm
¾–1¼ in

BERRY'S BOBTAIL SQUID
Euprymna berryi
F: Sepiolidae

Bobtail squid, such as this Indo-Pacific species, are small cuttlefish relatives. They have a pen, not cuttlebone, and the rounded body has lobed fins.



25–35 cm
10–14 in

BIGFIN REEF SQUID
Sepioteuthis lessoniana
F: Loliginidae

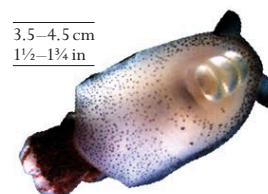
A pair of large fins makes this Indo-Pacific squid resemble a cuttlefish. It communicates by flashing light from special organs, which contain light-emitting bacteria.



30–45 cm
12–18 in

COMMON SQUID
Loligo vulgaris
F: Loliginidae

This commercially important squid is common in the N.E. Atlantic and Mediterranean. Like related species, it has prominent side fins.



3.5–4.5 cm
1½–1¾ in

RAM'S HORN SQUID
Spirula spirula
F: Spirulidae

This small, deep-ocean cephalopod contains a spiral gas-filled shell that acts as a float; the animal rises towards the ocean surface at night.